

Midpoint preservation on a quarter ball

MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball

theorem

Supplies both lens inclusions from matching ambient balls.

Statement

Let E, F be real normed spaces, $S \subseteq E$, $T \subseteq F$, and $f : S \rightarrow T$ a bijective isometry. Let $c \in S$ and $R > 0$ satisfy $B(c, R) \subseteq S$ and $B(f(c), R) \subseteq T$, where $B(c, R) = \{z : \|z - c\| < R\}$. For $x, y \in S \cap B(c, R/4)$,

$$f\left(\frac{x + y}{2}\right) = \frac{f(x) + f(y)}{2}.$$

Assumptions

Both ambient open-ball inclusions, positivity of R , and membership of both x, y in the quarter ball are required. The sets S, T need not themselves be open or convex.

Conclusion

The midpoint belongs to S , and the same isometry preserves that midpoint.

Proof route

Use radius $R/2$ for the source and target lenses. The quarter-ball bounds leave enough room for both lenses inside the given radius- R balls.

▼ Proof steps

1. The triangle inequality gives $\|x - y\| \leq \|x - c\| + \|y - c\| < R/2$, so $\|x - y\|/2 < R/4 < R/2$. Also

$$\left\| \frac{x + y}{2} - c \right\| \leq \frac{\|x - c\| + \|y - c\|}{2} < R/4 < R.$$

Thus the midpoint is a valid argument of f .

2. For $z \in \Lambda_{R/2}(x, y) = \overline{B}(x, R/2) \cap \overline{B}(y, R/2)$,

$$\|z - c\| \leq \|z - x\| + \|x - c\| < R/2 + R/4 < R.$$

Hence this entire lens lies in $B(c, R) \subseteq S$.

3. Isometry gives $\|f(x) - f(c)\| = \|x - c\| < R/4$. If $w \in \Lambda_{R/2}(f(x), f(y))$, then

$$\|w - f(c)\| \leq \|w - f(x)\| + \|f(x) - f(c)\| < R/2 + R/4 < R.$$

The target ball assumption now puts the entire target lens in T .

4. Apply Midpoint preservation under two lens inclusions to this f , the points x, y , and the radius $R/2$. Step 1 supplies its half-distance bound, and Steps 2–3 supply its two inclusions. Its output is exactly the claimed midpoint equation.

Main citations

- Exact theorem and proof
- Midpoint preservation under two lens inclusions

Lean source signature (exact)

```
theorem map_midpoint_of_mem_ball
  {s : Set E} {t : Set F} (f : s ≈i t) (c : s) {R : ℝ} (hR : 0 < R)
  (hsball : ball (c : E) R ⊆ s)
  (htball : ball ((f c : t) : F) R ⊆ t)
  (x y : s) (hx : (x : E) ∈ ball (c : E) (R / 4))
  (hy : (y : E) ∈ ball (c : E) (R / 4)) :
  ((f (midpoint ℝ (x : E) (y : E)),
    hsball (by
      rw [mem_ball]
      calc
        dist (midpoint ℝ (x : E) (y : E)) (c : E) ≤
          (dist (x : E) (c : E) + dist (y : E) (c : E)) / 2 := by
            simpa [dist_comm] using
              dist_midpoint_midpoint_le (x : E) (y : E) (c : E) (c : E)
          _ < R := by rw [mem_ball] at hx hy; linarith)) : t) : F) =
    midpoint ℝ ((f x : t) : F) ((f y : t) : F)
```

In the source	Mathematical meaning
$(f : s \approx_i t) (c : s)$	The distance-preserving bijection $f : S \rightarrow T$ and center $c \in S$, with $s \subseteq E, t \subseteq F$ in real normed spaces.
$\{R : \mathbb{R}\} (hR : 0 < R)$	The common radius $R > 0$.
$(hsball : \text{ball } (c : E) R \subseteq s)$	The open ball $B(c, R)$ lies in the source set.
$(htball : \text{ball } ((f c : t) : F) R \subseteq t)$	The open ball $B(f(c), R)$ lies in the target set; this is a separate hypothesis.
$(x y : s)$	Two points $x, y \in S$.
$(hx : (x : E) \in \text{ball } (c : E) (R / 4))$	The inequality $\ x - c\ < R/4$.
$(hy : (y : E) \in \text{ball } (c : E) (R / 4))$	The inequality $\ y - c\ < R/4$ for the other point.

In the source	Mathematical meaning
<code>hsball</code> (by	The following proof block verifies that $(x + y)/2 \in B(c, R) \subseteq S$ from the displayed distance estimate. It is not an extra assumption.
<code>midpoint ℝ ((f x : t) : F) ((f y : t) : F)</code>	The right side is $(f(x) + f(y))/2$. The entire equality says that applying f to the source midpoint gives this point in F .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

Accepted content SHA-256: `e19ce25a3ad10fdca363466e027beec33a87addbaa5acb304642b4fe1ed3caf`

Accepted source guide SHA-256: `bfc080efac19e4c51ee91abba08bb204ed986f22a088cd8eef3c8d572fa2d97c`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`